

SK. MD. ASIF NEWAZ

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RESEARCH INTERESTS

Machine Learning · Multi-Agent Systems · Multimodal Integration · Agentic AI · Health-Informatics · Big Data-Analytics · LLM · RAG · Explainable AI · Medical Image Analysis · Contrastive Learning ·

EDUCATION

B.Sc. in Computer Science and Engineering September 2021 – April 2026
American International University-Bangladesh (AIUB), Dhaka, Bangladesh
CGPA: 3.89/4.00
Relevant Coursework: Machine Learning (Python), Introduction to Data Science (R), Research Methodology, Artificial Intelligence and Expert System (Python), Computational Statistics and Probability (MATLAB).
Thesis: “Binary Retrieval-Augmented Generation (BinaryRAG): Contrastive Learning and Multimodal Integration for Enhanced Monkeypox Detection”

ENGLISH LANGUAGE PROFICIENCY

IELTS Academic 28 August 2026
Overall Band Score: **7.5** (CEFR Level C1)
Listening: 8.5 Reading: 7.5 Writing: 6.5 Speaking: 7.5

RESEARCH EXPERIENCE

Research Assistant April 2026 – Present
Elite Research Lab PLC. Queens, NY, USA (Remote)
Supervisor: Md Kishor Morol

- Leading a government-scale RAG tutoring system over 132 NCTB textbooks in Bengali, Arabic, and English; audited and re-converted the entire corpus from source PDFs, discovering vector-outlined text and evaluating generative vs. classical OCR approaches.
- Conducting a comprehensive audit of publicly available PCOS datasets (tabular and image) for data quality, leakage, duplication, and shortcut learning; identified criterion feature leakage in a high-visibility Kaggle dataset (23.8K downloads) and contributed to a verification atlas.
- Developing a multimodal deep learning pipeline for PCOS detection integrating ultrasound images with patient-matched clinical data, with a focus on robust fusion and real-world validation.

Research Intern March 2025 – June 2025
Applied Intelligence and Informatics Lab (AIIL). Nottingham, United Kingdom (Remote)
Supervisor: Dr. Md. Saef Ullah Miah

- Conducted in-depth literature reviews and developed/fine-tuned large language models, including Mistral-7B-Instruct-v0.3, for culturally aware sales negotiation systems in commercial contexts.
- Compiled datasets, performed model selection and comparison across architectures (Flan-T5, Gemma-2B, LLaMA-3B), and addressed technical implementation challenges in resource-constrained environments.
- Managed editorial works and indexing for ICONIP2024-CCIP conference papers via ProConf platform.

PUBLICATIONS

Journal Submission (Thesis – Waiting for funding approval)

Binary Retrieval-Augmented Generation (BinaryRAG): Contrastive Learning and Multimodal Integration for Enhanced Monkeypox Detection 2026

- Novel Binary-RAG framework combining CLIP visual embeddings + SentenceTransformer clinical metadata with contrastive multi-head attention

International Conference Papers (All Accepted/Published)

Chlorella vulgaris Image Classification Using EfficientNetV2-S, ResNet50 and InceptionResNetV2 with Grad-CAM Analysis (ICIIR 2025)

- 82.65% accuracy (EfficientNetV2-S) and mIoU 0.884 (InceptionResNetV2) on CIDACC dataset

Improving Diagnosis and Treatment Planning for Heart Diseases Using Virtual Reality Integrated with Explainable AI [Best Paper Award] (ICIIR 2025)

- Integration of Virtual Reality (VR) with Explainable AI (XAI) to substantially improve diagnostic accuracy, treatment planning, and medical education outcomes for cardiovascular diseases

Predicting Type 1 Diabetes Using Machine Learning (ICIDA 2024)

- Compared RF, Gradient Boosting, SVM, Logistic Regression for early risk prediction

Breast Cancer Detection Using Machine Learning Algorithms (ICIDA 2024)

- Hybrid ensemble (SVM, KNN, RF, AdaBoost, PCA, GB, DT) on benchmark datasets

Data-Driven Diagnosis of Monkeypox: Insights from Machine Learning Models [Best Paper Award] (TEHI 2024)

- Evaluated SVM, DT, RF, LR, XGBoost, LightGBM; 4–8% gain via targeted data cleaning

Conference Papers (Accepted)

The Follicle Count Paradox: Measuring Diagnostic Leakage in AI-Driven PCOS Prediction (ICCA 2026)

- Identified Rotterdam criterion leakage of the most popular PCOS dataset on Kaggle with 23.9k downloads in terms of ML based binary classification.

TECHNICAL SKILLS

Language & Frameworks: Python, R, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, CLIP, Sentence-Transformers, LangChain

Deep Learning: CNNs, EfficientNet, ResNet, Inception, Vision Transformers, Contrastive Learning, Multi-head Attention, Reranker

Tools & Platforms: Jupyter, Git, LaTeX, Grad-CAM, Docker, n8n, PgVector, SPSS, Linux, AWS/GCP (basic), LM Studio, Ollama, UnSloth

ACADEMIC ACHIEVEMENTS

- Dean's List Honors (all semesters)
- Academic Scholarship for Excellence
- Recipient of Magna Cum Laude (Silver Medal) for overall academic performance
- Acquired two best paper awards at separate international conferences [TEHI 2024, ICIIR 2025]

REFERENCES

Dr. Md. Saef Ullah Miah

Research Lead, Applied Intelligence and Informatics Lab
Director, FST, American International University-Bangladesh
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Additional references available upon request.

Md. Kishor Morol

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